

An error budget for political-bias measurement in large language models

Abstract

Large language models are widely reported to lean left of centre, replicated across dozens of studies. We show this is substantially a property of measurement. Administering the Political Compass and 8Values to 19 models, 60 times each (2,280 administrations), we decompose a measured political position into variance from model identity, instrument choice, and trial noise. Instrument and axis choice accounts for 52% of variance (95% CI [35%, 72%]) and model identity for only 13% ([0.02%, 22%]): which questionnaire an auditor picks moves a score four times more than which model is audited. The two instruments also fail convergent validity: $r = -0.26$ for shared constructs versus $r = +0.52$ for distinct ones on the same instrument. Crossing two further instruments, SupplyValues and Pew 2026 Political Typology, into the same model collapses the pooled model-identity share to 1.89%. Three additional factors, asker identity, instrument contamination, and response format, each explain more variance in their factorial designs than model identity explains under the standard protocol. Reported bias magnitudes are therefore not comparable across studies using different instruments, and every factor examined explains more of a score than the model producing it does. We propose a minimum reporting standard for such audits.

Large language models now mediate political information for hundreds of millions of people, and a substantial literature reports they lean left of centre, replicated across model families, providers, and languages (Rozado 2024; Motoki et al. 2024; Hartmann et al. 2023; Feng et al. 2023; Buyl et al. 2025; Peng et al. 2026; Sakhawat et al. 2026). Each assumption behind this measurement has since been challenged in isolation: models accommodate the political identity they infer for the asker (Törnberg and Schimmel 2026); the instruments themselves appear in pretraining data (Bianchi et al. 2026); forced-choice framing produces answers models do not give when responding freely (Röttger et al. 2024). None has been quantified against the others, so the field has no estimate of how much of a reported score is the model and how much is the apparatus. The closest existing study administers three instruments to 26 models and obtains a comparable model-versus-prompt split, but measures each cell once, so trial noise cannot be separated from the instrument term (Sakhawat et al. 2026); repeating each cell and scoring against each instrument’s own authoritative engine, rather than an approximate formula, is what makes the noise term below possible.

We supply that estimate. Administering the Political Compass and 8Values to 19 models from 11 organizations, spanning open and closed weights and roughly two orders of magnitude in price, 60 times each (2,280 administrations), we decompose a measured political position into variance from model identity, instrument and axis choice, and residual trial noise, each with a 95% CI from a cluster bootstrap over the 19 models. Pooled across instruments, instrument and axis choice accounts for 52.2% of variance ([35.0%, 71.9%]) and model identity for only 13.0% ([0.02%, 21.9%]) (Fig. 1a; Table 1): which questionnaire an auditor picks moves a reported score roughly four times more than which model is audited, and residual trial-to-trial noise (34.8%, [18.9%, 49.3%]) exceeds

the model term. Conditional on a fixed axis, where the between-instrument term is removed by construction, model identity accounts for 66.6% of remaining variance (Fig. 1b): models are clearly separable once the instrument is held constant, and both figures are correct, but they answer different questions, and only the first governs comparisons across studies using different instruments.

The two instruments are not interchangeable measures of one underlying position. A multi-trait multi-method analysis finds convergent validity—the same construct measured by both instruments—averaging $r = -0.26$ across the 19 models, against $r = +0.52$ for discriminant validity (different constructs, same instrument); the Campbell–Fiske ordering, which requires the first to exceed the second, is inverted. Crossing two further instruments, SupplyValues and Pew Research Center’s 2026 Political Typology, scored by driving Pew’s live public quiz and reading off its own classification rather than reimplementing its unpublished clustering procedure, into the same crossed random-effects model collapses the pooled model-identity share further, to 1.89% (8,640 administrations; Table 1).¹ The collapse is driven by SupplyValues actively disagreeing with Political Compass on model ranking ($\rho = -0.49$), not by Pew’s coarser ordinal scale diluting the signal: adding SupplyValues alone drops the model share to 0.49%, adding Pew alone only to 8.21%.

Three further apparatus components, each demonstrated only in isolation in the prior literature, are larger than the model term when measured directly on the same models (Table 1). Framing the asker’s identity —conservative, progressive, apolitical, journalist, academic researcher, or none—explains more variance ($\eta^2 = 0.79$, [0.65, 0.93]) than model identity does under the field’s standard unattributed condition ($\eta^2 = 0.56$ –0.73); accommodation is asymmetric, moving scores 2.9–6.7 times further from baseline for a conservative framing than a progressive one. Sense-inverting instrument items with the scoring key flipped produces a Contamination Index averaging 0.046 ([0.038, 0.055]) of the instrument’s own range, with 17 of 18 models shifting on at least one axis after correction, and a companion memorisation probe (16 of 18 models parsed) finds a 16.6% verbatim item-completion rate. Free-text elicitation, judged onto the same scale, versus a forced label explains more variance (39%, [0.02, 0.74]) than model identity within the same design (23%, [0.04, 0.52]).

None of these three factors is folded into the pooled budget above, since each was collected as its own factorial against the field’s standard baseline rather than crossed with instrument simultaneously; doing so is the natural next step and would likely push the already-small model share down further still. That is the direction every factor examined has pushed, and none has pushed the other way.

We are not arguing models are politically neutral, nor that any specific published estimate is wrong, nor that a questionnaire score predicts real-world behaviour: a separate, dual-instrument study finds voting-advice questionnaires and concrete referendum votes disagree (Barnettler 2026), a behavioural-validity question this design does not test. Within a single study using one instrument, model identity is clearly measurable and rank orderings are meaningful. The failure is transportability: published magnitudes are not comparable across studies using different instruments, askers, or elicitation formats, and every apparatus component we examined explains more of a reported score than the model producing it does.

We propose that a political-bias audit report, at minimum: the instrument, named; the asker condition, including the absence of one; the repetition count and dispersion, in the instrument’s

¹Automating Pew’s live quiz at this scale (18 models $\times$ 10 trials) is a distinct question from the survey items’ own reuse terms and plausibly exceeds ordinary personal use of that interactive tool; we disclose this explicitly (web appendix, Ethics Declarations).

own units; a variance decomposition across whatever was varied; a contamination check; and item-level, not only axis-level, results. Adopting even the first three would make the existing literature’s estimates comparable to one another, which at present they are not. Full methods, the complete four-instrument crossing, and item-response-theory, prompt-robustness, and open-ended-generation checks that corroborate these results are in the web appendix.

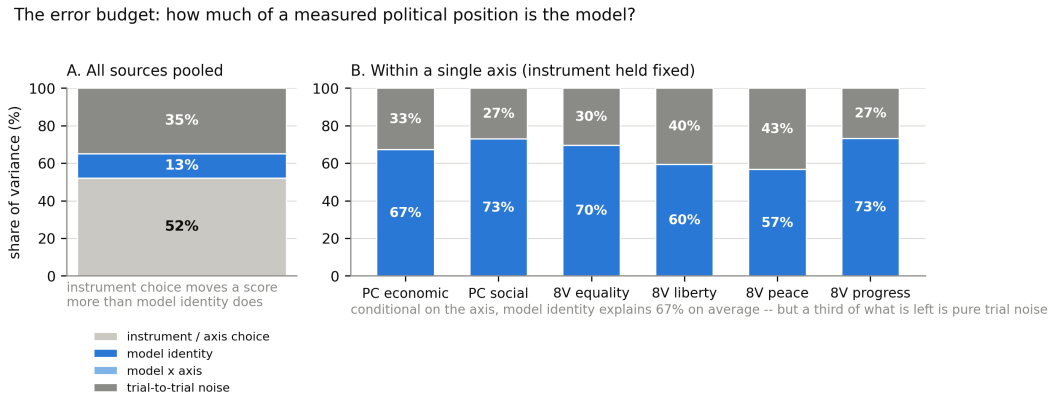


Figure 1: **The error budget.** Variance components of a measured political position, from a crossed random-effects model on 2,280 administrations. **(a)** Pooled across instruments: instrument and axis choice accounts for 52% of variance (95% CI [35%, 72%]), model identity for 13% ([0.02%, 22%]), residual trial noise for 35% ([19%, 49%]); cluster bootstrap over 19 models, 2,000 resamples. **(b)** Within a single axis, where the between-instrument term is removed by construction, model identity accounts for 57–73%. Panel (a) governs comparison across studies using different instruments; panel (b) governs comparison within one study.

Table 1: Every measurement factor examined explains more variance than model identity does under the field’s standard protocol. Model identity’s share is reported under whichever baseline each factor’s own design used; intervals are cluster bootstraps over models. Full designs in the web appendix.

Factor	Model identity	Factor itself	<i>n</i>
Instrument (2 instruments)	13.0% [0.02, 21.9]	52.2% [35.0, 71.9]	2,280
Instrument (4 instruments)	1.89% [0.0, 6.0]	—	8,640
Asker identity	η^2 0.56–0.73 (unattrib.)	η^2 0.79 [0.65, 0.93]	2,160
Response format	η^2 0.23 [0.04, 0.52]	η^2 0.39 [0.02, 0.74]	4,320
Instrument contamination	—	CI 0.046 [0.038, 0.055]; 17/18 models	1,080

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